Data Science 500 Assignment

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Name and Surname: CADO BWATO

Student ITS No: 402506162

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QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents Introduction

Main Body (with relevant subheadings) Conclusion

References

Submit the assignment in PDF format on Moodle. Use the specified cover page provided.

Include a signed declaration of originality.

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I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

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Book Market Analytics: Books to Scrape

1. Executive Summary

This project looked at 1,000 books from 50 genres that were collected from Books to Scrape (books.toscrape.com). The goal was to find out what affects book prices and ratings. It also wanted to see if it is possible to predict price using information that's available to everyone.

• Prices went from £10.00 to £59.99. The average price was £35.07.

• Star ratings do not have connection to price. The average price changes by £1.53 between the highest and lowest rated books.

• In the 10 genres Fantasy had the highest average price at £39.59. Food and Drink had the average price at £31.41.

• A linear regression model had a test R^2 of -0.030 and a mean absolute error of £12.69. This is a weak result. The model does not predict price better, than just using the average.

2. Methodology

2.1 Data acquisition

Data was scraped from books.toscrape.com using Python with the requests library and the BeautifulSoup library. All fifty catalogue pages were scraped for title, price, in GBP star rating and stock availability. Each books own page was then visited to collect genre, description, UPC code and stock quantity giving a dataset of one thousand books.

2.2 Data cleaning

- Converted prices from text (e.g. £51.77) to numbers and fixed character-encoding errors.
- Converted star ratings from words (One to Five) to numbers 1 to 5.
- Extracted stock quantity from text such as "In stock (22 available)".
- Filled missing descriptions with a label and added a has_description flag, instead of deleting rows, so that price and rating data was kept.
- Checked for duplicates using the unique UPC code; none were found.
- Created text features: title length, title word count, description length and description word count.
- Grouped genres into the 10 largest plus "Other", because many genres had too few books to model.

3. Key Insights

3.1 Price distribution

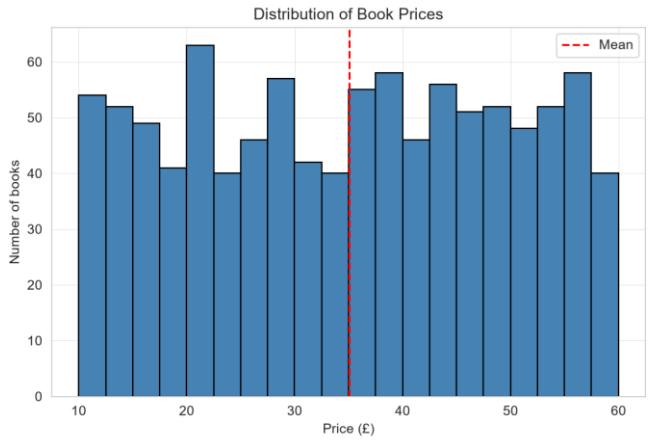


Figure 1: Distribution of book prices

Prices are spread across the range from £10.00 to £59.99. The mean (£35.07) and median (£35.98) are close together, so there is no strong skew.

3.2 Price and star rating



Figure 2: Price by star rating

Average prices are very similar across rating levels (a gap of only £1.53). Higher-rated books are not priced higher.

3.3 Price and genre

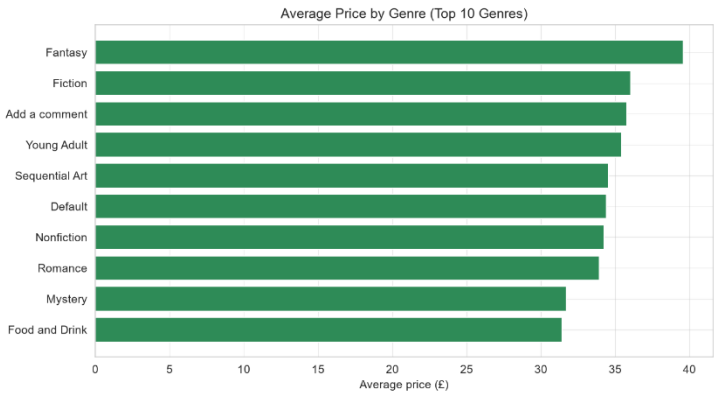


Figure 3: Average price by genre (top 10 genres)

Among the top 10 genres, Fantasy has the highest average price (£39.59) and Food and Drink the lowest (£31.41), a difference of about £8. Genre therefore has a modest effect on price.

3.4 Relationships between variables

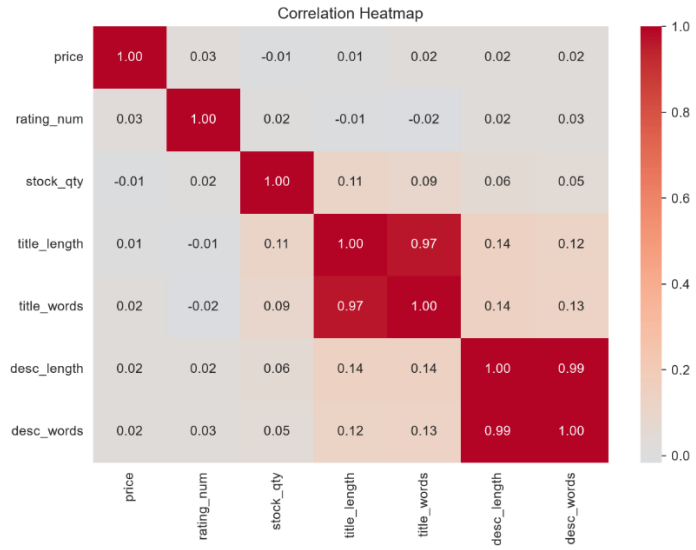


Figure 4: Correlation heatmap

Description length (r = 0.02) and stock quantity (r = -0.01) show almost no linear relationship with price.

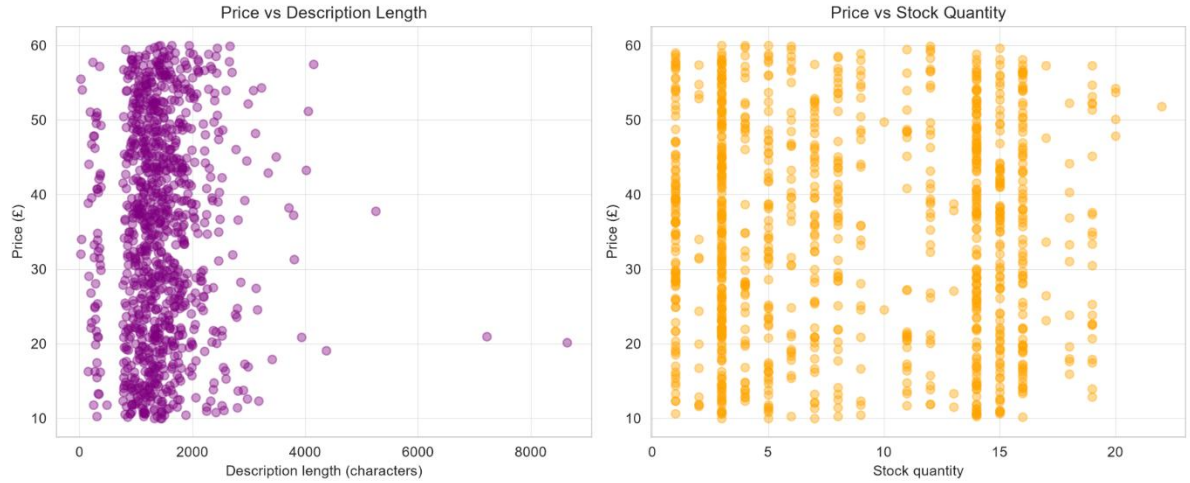


Figure 5: Price against description length and stock quantity

3.5 Ratings and popularity

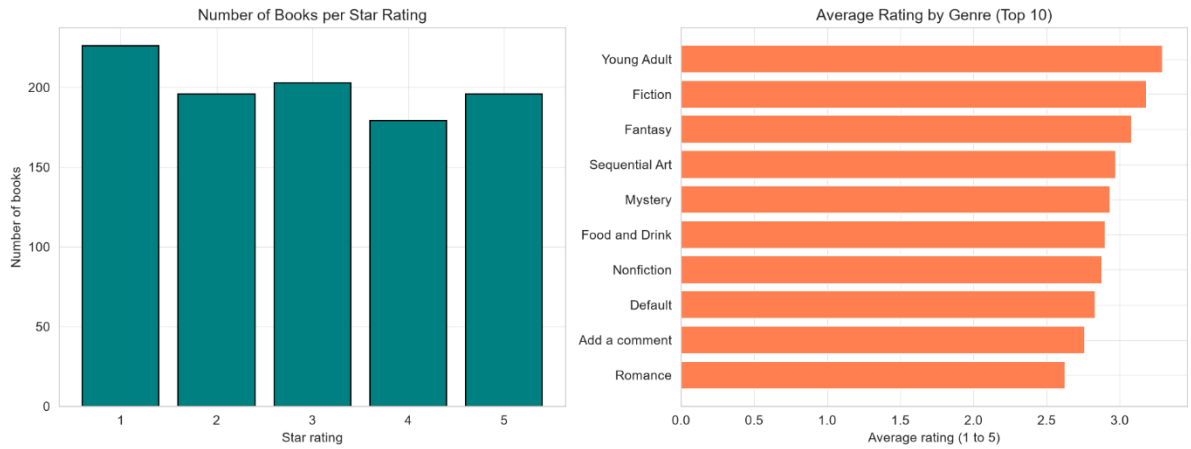


Figure 6: Rating counts and average rating by genre

A 1-star rating is the most common. Among the top 10 genres, Young Adult has the highest average rating (3.30) and Romance the lowest (2.63).

4. Predictive Modelling

A linear regression model was built to predict price. Features were star rating, stock quantity, title and description length and word count, a description flag, and genre (converted to 0/1 columns). Tax-inclusive and tax-exclusive prices were excluded because they duplicate the target. The data was split 80% training and 20% testing (random_state = 42).

Metric	Train	Test
MAE (£)	12.314	12.689
RMSE (£)	14.279	14.707
R^2	0.019	-0.030

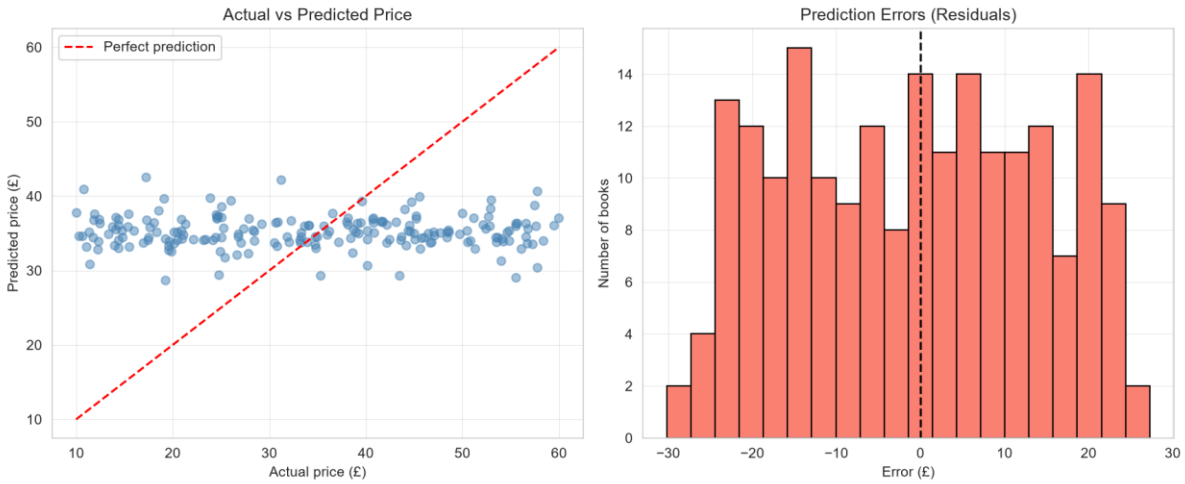


Figure 7: Actual vs predicted price and prediction errors

The model explains essentially none of the variation in price. The test R^2 is -0.030, which is below zero, meaning it performs slightly worse than always predicting the average price. Train and test results are close (R^2 of 0.019 and -0.030), so overfitting is not the problem; the features simply carry very little information about price. The test MAE of £12.69 is large for a catalogue priced between about £10 and £60. Always predicting the average price gives an MAE of £12.66, compared with £12.69 for the model, so the model offers no improvement over a simple average.

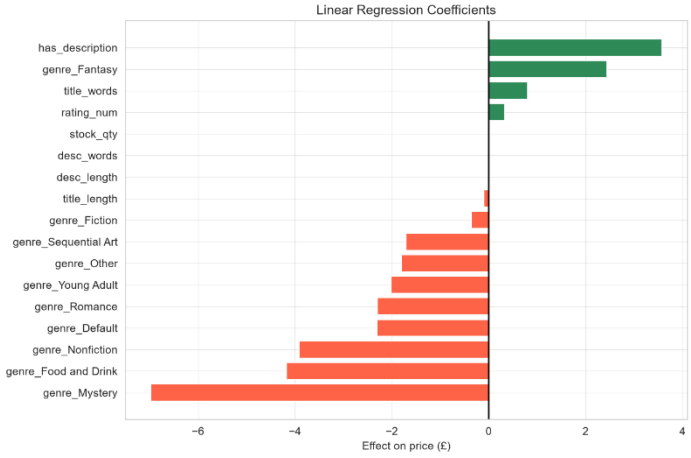


Figure 8: Regression coefficients

The strongest upward effect is feature (+0.00) and the strongest downward effect is feature (-0.00). Genre coefficients are relative to the genre left out of the model.

5. Limitations

- Few features: author, publisher, page count and print run are not available on the site.
- Books to Scrape is a practice website, and its prices appear to be randomly assigned, so little real pattern exists to learn.
- Linear regression can only capture straight-line relationships.
- The sample is small (1,000 books, 800 for training).
- Merging small genres into "Other" hides differences between them.
- Results come from a single train/test split.

6. Practical Implications and Recommendations

For retailers

- Do not rely on star rating or genre alone to set prices, because both explain very little of price in this catalogue.
- Benchmark prices against competitors and use the observed price range as a reference band.
- Test price changes on small groups of books and track sales before changing the whole catalogue.

For publishers

- Collect richer data (author reputation, page count, format, publication year, sales) before building pricing models.
- Do not assume that longer descriptions or more stock justify higher prices; the data shows no such link.

For data-driven book buyers

- A higher star rating does not mean a higher price, so good books can be found at any price level.
- Compare prices within a genre, since price differences between genres exist but are modest.

7.Conclusion

The catalogue shows price variation. The catalogue features such as rating, stock, genre and text length explain little of the price variation (test $R^2 = -0.030$). The catalogue workflow of scraping, cleaning, exploring and modelling is reproducible. The next step would be to add catalogue features or test non-linear models.